Ευφυείς Πράκτορες και Ρομποτικά Συστήματα

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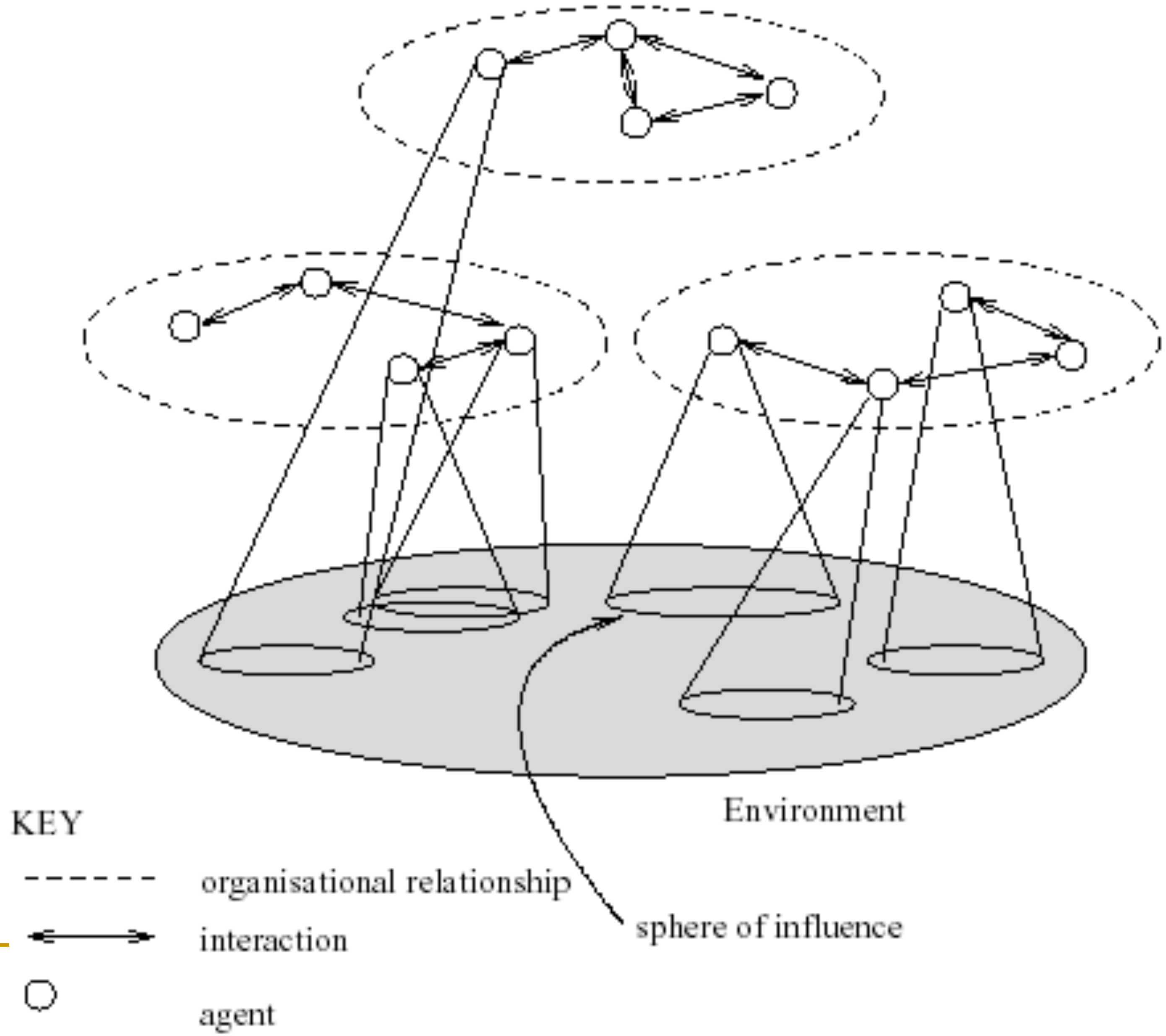
MULTIAGENT INTERACTIONS



MultiAgent Systems

Thus a multiagent system contains a number of agents…

- …which interact through communication…
- …are able to act in an environment…
- …have different “spheres of influence” (which may coincide)…
- …will be linked by other (organizational) relationships

Self-interested agents

- What does it mean to say that an agent is self-interested?
  - not that they want to harm other agents
  - not that they only care about things that benefit them
  - that the agent has its own description of states of the world that it likes, and that its actions are motivated by this description

How to model agents' interests?

- Utility theory:
  - quantifies degree of preference across alternatives
  - understand the impact of uncertainty on these preferences
  - utility function : a mapping from states of the world to real numbers, indicating the agent's level of happiness with that state of the world
  - Decision-theoretic rationality : take actions to maximize expected utility.

Friends and enemies

- Alice has three options: club (c), movie (m), watching a video at home (h)
- On her own, her utility for these three outcomes is 100 for c, 50 for m and 50 for h
- However, Alice also cares about Bob (who she hates) and Carol (who she likes)
- Bob is at the club 60% of the time, and at the movies otherwise
- Carol is at the movies 75% of the time, and at the club otherwise
- If Alice runs into Bob at the movies, she suffers disutility of 40; if she sees him at the club she suffers disutility of 90.
- If Alice sees Carol, she enjoys whatever activity she's doing 1.5 times as much as she would have enjoyed it otherwise (taking into account the possible disutility caused by Bob)
- What Should Alice do?

What activity should Alice choose?

	$B = c$	$B = m$
$C = c$	15	150
$C = m$	10	100

$A = c$

	$B = c$	$B = m$
$C = c$	50	10
$C = m$	75	15

$A = m$

What activity should Alice choose?

$A = c$

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$A = m$

	$B = c$	$B = m$
$C = c$	50	10
$C = m$	75	15

- Alice's expected utility for c:

0.25(0.6 *15 + 0.4 *150) + 0.75(0.6 *10 + 0.4 *100) = 51.75:

- Alice's expected utility for m:

0.25(0.6*50 + 0.4*10) + 0.75(0.6*75 + 0.4*15) = 46.75:

- Alice's expected utility for h: 50.

Alice prefers to go to the club (though Bob is often there and Carol rarely is), and prefers staying home to going to the movies (though Bob is usually not at the movies and Carol almost always is).

Why utility?

- Why would anyone argue with the idea that an agent's preferences could be described using a utility function as we just did?
- why should a single-dimensional function be enough to explain preferences over an arbitrarily complicated set of alternatives?
- Why should an agent's response to uncertainty be captured purely by the expected value of his utility function?
- It turns out that the claim that an agent has a utility function is substantive.

Preferences over outcomes

If o_1 and o_2 are outcomes

- $o_1 \succeq o_2$ means o_1 is at least as desirable as o_2.
  - read this as "the agent weakly prefers o_1 to o_2"
- $o_1 \sim o_2$ means $o_1 \succeq o_2$ and $o_2 \succeq o_1$.
  - read this as "the agent is indifferent between o_1 and o_2."
- $o_1 \succ o_2$ means $o_1 \succeq o_2$ and $o_2 \not\succeq o_1$
  - read this as "the agent strictly prefers o_1 to o_2"

Lotteries

- An agent may not know the outcomes of his actions, but may instead only have a probability distribution over the outcomes.

Definition (lottery)

A lottery is a probability distribution over outcomes. It is written

$$[p_1 : o_1, p_2 : o_2, \ldots, p_k : o_k]$$

where the o_i are outcomes and $p_i > 0$ such that

$$\sum_i p_i = 1$$

- The lottery species that outcome o_i occurs with probability p_i.
- We will consider lotteries to be outcomes.

Preference Axioms

Definition (Completeness)

A preference relationship must be defined between every pair of outcomes:

$$\forall o_1 \forall o_2 \; o_1 \succeq o_2 \text{ or } o_2 \succeq o_1$$

Definition (Transitivity)

Preferences must be transitive:

$$\text{if } o_1 \succeq o_2 \text{ and } o_2 \succeq o_3 \text{ then } o_1 \succeq o_3$$

Definition (Monotonicity)

An agent prefers a larger chance of getting a better outcome to a smaller chance:

- If $o_1 \succ o_2$ and $p > q$ then

$$[p : o_1, 1 - p : o_2] \succ [q : o_1, 1 - q : o_2]$$

Definition (Substitutability)

If $o_1 \sim o_2$ then for all sequences of one or more outcomes $o_3, \ldots, o_k$ and sets of probabilities $p, p_3, \ldots, p_k$ for which $p + \sum_{i=3}^{k} p_i = 1$,
$[p : o_1, p_3 : o_3, \ldots, p_k : o_k] \sim [p : o_2, p_3 : o_3, \ldots, p_k : o_k]$.

Preference Axioms

Let $P_\ell(o_i)$ denote the probability that outcome o_i is selected by lottery ℓ. For example, if $\ell = [0.3 : o_1; 0.7 : [0.8 : o_2; 0.2 : o_1]]$ then $P_\ell(o_1) = 0.44$ and $P_\ell(o_3) = 0$.

Definition (Decomposability ("no fun in gambling"))

If $\forall o_i \in O$, $P_{\ell_1}(o_i) = P_{\ell_2}(o_i)$ then $\ell_1 \sim \ell_2$.

Definition (Continuity)

Suppose $o_1 \succ o_2$ and $o_2 \succ o_3$, then there exists a $p \in [0, 1]$ such that $o_2 \sim [p : o_1, 1 - p : o_3]$.

Preferences and utility functions

Theorem (von Neumann and Morgenstern, 1944)

If an agent's preference relation satisfies the axioms Completeness, Transitivity, Decomposability, Substitutability, Monotonicity and Continuity then there exists a function $u : O \rightarrow [0, 1]$ with the properties that:

1. $u(o_1) \geq u(o_2)$ *iff the agent prefers* o_1 *to* o_2*; and*
2. *when faced about uncertainty about which outcomes he will receive, the agent prefers outcomes that maximize the expected value of* u.

Utilities and Preferences

- Assume we have just two agents: $Ag = \{i, j\}$
- Agents are assumed to be *self-interested*: they *have preferences over how the environment is*
- Assume $\Omega = \{\omega_1, \omega_2, \ldots\}$ is the set of "outcomes" that agents have preferences over
- We capture preferences by *utility functions*:

$$u_i = \Omega \to \mathbb{R}$$
$$u_j = \Omega \to \mathbb{R}$$

- Utility functions lead to *preference orderings* over outcomes:

$$\omega \geq \omega' \text{ means } u_i(\omega) \geq u_i(\omega')$$
$$\omega > \omega' \text{ means } u_i(\omega) > u_i(\omega')$$

Multiagent Encounters

- We need a model of the environment in which these agents will act…
  - agents simultaneously choose an action to perform, and as a result of the actions they select, an outcome in Ω will result
  - the *actual* outcome depends on the *combination* of actions
  - assume each agent has just two possible actions that it can perform, *C* ("cooperate") and *D* ("defect")
- Environment behavior given by *state transformer function*:

$$\tau : \underbrace{Ac}_{\text{agent } i\text{'s action}} \times \underbrace{Ac}_{\text{agent } j\text{'s action}} \rightarrow \Omega$$

Multiagent Encounters

- Here is a state transformer function:

  $$\tau(D, D) = \omega_1 \quad \tau(D, C) = \omega_2 \quad \tau(C, D) = \omega_3 \quad \tau(C, C) = \omega_4$$

  (This environment is sensitive to actions of both agents.)
- Here is another:

  $$\tau(D, D) = \omega_1 \quad \tau(D, C) = \omega_1 \quad \tau(C, D) = \omega_1 \quad \tau(C, C) = \omega_1$$

  (Neither agent has any influence in this environment.)
- And here is another:

  $$\tau(D, D) = \omega_1 \quad \tau(D, C) = \omega_2 \quad \tau(C, D) = \omega_1 \quad \tau(C, C) = \omega_2$$

  (This environment is controlled by *j*.)

Rational Action

- Suppose we have the case where *both* agents can influence the outcome, and they have utility functions as follows:
$$u_i(\omega_1) = 1 \quad u_i(\omega_2) = 1 \quad u_i(\omega_3) = 4 \quad u_i(\omega_4) = 4$$
$$u_j(\omega_1) = 1 \quad u_j(\omega_2) = 4 \quad u_j(\omega_3) = 1 \quad u_j(\omega_4) = 4$$
- With a bit of abuse of notation:
$$u_i(D, D) = 1 \quad u_i(D, C) = 1 \quad u_i(C, D) = 4 \quad u_i(C, C) = 4$$
$$u_j(D, D) = 1 \quad u_j(D, C) = 4 \quad u_j(C, D) = 1 \quad u_j(C, C) = 4$$
- Then agent i's preferences are:
$$C, C \succeq_i C, D \quad \succ_i \quad D, C \succeq_i D, D$$
- "*C*" is the *rational choice* for i.
(Because i prefers all outcomes that arise through *C* over all outcomes that arise through *D*.)

Introduction to Game Theory

Non-Cooperative Game Theory

- What is it?

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Non-Cooperative Game Theory

- What is it?
  - mathematical study of interaction between rational, self-interested agents
- Why is it called non-cooperative?
  - while it's most interested in situations where agents' interests conflict, it's not restricted to these settings
  - the key is that the individual is the basic modeling unit, and that individuals pursue their own interests
    - cooperative/coalitional game theory has teams as the central unit, rather than agents

TCP Backoff Game

TCP Backoff Game

Should you send your packets using correctly-implemented TCP (which has a “backoff" mechanism) or using a defective implementation (which doesn’t)?

Consider this situation as a two-player game:

- both use a correct implementation: both get 1 ms delay
- one correct, one defective: 4 ms delay for correct, 0 ms for defective
- both defective: both get a 3 ms delay.

TCP Backoff Game

- Consider this situation as a two-player game:
  - both use a correct implementation: both get 1 ms delay
  - one correct, one defective: 4 ms delay for correct, 0 ms for defective
  - both defective: both get a 3 ms delay.
- Questions:
  - What action should a player of the game take?
  - Would all users behave the same in this scenario?
  - What global patterns of behaviour should the system designer expect?
  - Under what changes to the delay numbers would behavior be the same?
  - What effect would communication have?
  - Repetitions? (finite? infinite?)
  - Does it matter if I believe that my opponent is rational ?

Games - Definition

- Finite, n-person game: $\langle N, A, u\rangle$:
  - N is a finite set of n players, indexed by i
  - $A = A_1 \times \ldots \times A_n$, where A_i is the action set for player i
    - $a \in A$ is an action profile, and so A is the space of action profiles
  - $u = \langle u_1, \ldots, u_n\rangle$, a utility function for each player, where $u_i : A \mapsto \mathbb{R}$

Note the implicit assumption that *O*=*A*

Payoff matrices

- Writing a 2-player game as a matrix:
  - row player is player 1, column player is player 2
  - rows are actions $a \in A_1$, columns are $a' \in A_2$
  - cells are outcomes, written as a tuple of utility values for each player

Games in Matrix Form

Here's the TCP Backoff Game written as a matrix ("normal form").

	C	D
C	$-1, -1$	$-4, 0$
D	$0, -4$	$-3, -3$

Games in Matrix Form

Prisoner's dilemma is any game

	C	D
C	a, a	b, c
D	c, b	d, d

with $c > a > d > b$.

Games of Pure Competition

Players have exactly opposed interests

- There must be precisely two players (otherwise they can't have exactly opposed interests)
- For all action profiles $a \in A$, $u_1(a) + u_2(a) = c$ for some constant c
    - Special case: zero sum
- Thus, we only need to store a utility function for one player
    - in a sense, it's a one-player game

Games of Pure Competition

One player wants to match; the other wants to mismatch.

	Heads	Tails
Heads	1	-1
Tails	-1	1

Games of Pure Competition

Generalized matching pennies.

	Rock	Paper	Scissors
Rock	0	−1	1
Paper	1	0	−1
Scissors	−1	1	0

...Believe it or not, there's an annual international competition for this game!

Games of Competition

Players have exactly the same interests.

- no conflict: all players want the same things
- $\forall a \in A, \forall i, j,\ u_i(a) = u_j(a)$
- we often write such games with a single payoff per cell
- why are such games "noncooperative"?

Games mixing Cooperation and Competition

The most interesting games combine elements of cooperation *and* competition.

	B	F
B	$2, 1$	$0, 0$
F	$0, 0$	$1, 2$

Analyzing Games

- We've defined some canonical games, and thought about how to play them. Now let's examine the games from the outside

- From the point of view of an outside observer, can some outcomes of a game be said to be better than others?

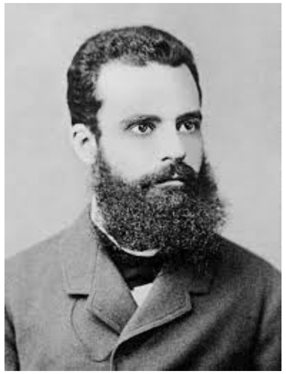
Analyzing Games

- We've defined some canonical games, and thought about how to play them. Now let's examine the games from the outside

- From the point of view of an outside observer, can some outcomes of a game be said to be better than others?
    - we have no way of saying that one agent's interests are more important than another's
    - intuition: imagine trying to find the revenue-maximizing outcome when you don't know what currency has been used to express each agent's payoff

- Are there situations where we can still prefer one outcome to another?

Pareto Optimality



- Idea: sometimes, one outcome o is at least as good for every agent as another outcome o', and there is some agent who strictly prefers o to o'
  - in this case, it seems reasonable to say that o is better than o'
  - we say that o Pareto-dominates o'.

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can a game have more than one Pareto-optimal outcome?

does every game have at least one Pareto-optimal outcome?

Pareto Optimality: Examples

	C	D
C	$-1, -1$	$-4, 0$
D	$0, -4$	$-3, -3$

	Left	Right
Left	1	0
Right	0	1

	B	F
B	$2, 1$	$0, 0$
F	$0, 0$	$1, 2$

	Heads	Tails
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Best Response

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- Let $a_{-i} = \langle a_1, \ldots, a_{i-1}, a_{i+1}, \ldots, a_n \rangle$.
  - now $a = (a_{-i}, a_i)$

- Best response: $a_i^* \in BR(a_{-i})$ iff
  $\forall a_i \in A_i,\ u_i(a_i^*, a_{-i}) \geq u_i(a_i, a_{-i})$

Nash Equilibrium

- Now let's return to the setting where no agent knows anything about what the others will do
- What can we say about which actions will occur?

- Idea: look for stable action profiles.
- $a = \langle a_1, \ldots, a_n \rangle$ is a ("pure strategy") Nash equilibrium iff $\forall i,\ a_i \in BR(a_{-i})$.

Nash Equilibrium: Examples

	C	D
C	$-1, -1$	$-4, 0$
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	Left	Right
Left	1	0
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	B	F
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Nash Equilibrium: Examples

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	B	F
B	2, 1	0, 0
F	0, 0	1, 2

	Heads	Tails
Heads	1	-1
Tails	-1	1

The paradox of Prisoner's dilemma: the Nash equilibrium is the only non-Pareto-optimal outcome!

Mixed Strategies

- It would be a pretty bad idea to play any deterministic strategy in matching pennies
- Idea: confuse the opponent by playing randomly
- Define a strategy s_i for agent i as any probability distribution over the actions A_i.
  - pure strategy: only one action is played with positive probability
  - mixed strategy: more than one action is played with positive probability
    - these actions are called the support of the mixed strategy
- Let the set of all strategies for i be S_i
- Let the set of all strategy profiles be $S = S_1 \times \ldots \times S_n$.

Utility under Mixed Strategies

- What is your payoff if all the players follow mixed strategy profile $s \in S$?
    - We can't just read this number from the game matrix anymore: we won't always end up in the same cell

Utility under Mixed Strategies

- What is your payoff if all the players follow mixed strategy profile $s \in S$?
  - We can't just read this number from the game matrix anymore: we won't always end up in the same cell

- Instead, use the idea of expected utility from decision theory:

$$u_i(s) = \sum_{a \in A} u_i(a) Pr(a|s)$$

$$Pr(a|s) = \prod_{j \in N} s_j(a_j)$$

Best Response and Nash Equilibrium

Our definitions of best response and Nash equilibrium generalize from actions to strategies.

- Best response:
  - $s_i^* \in BR(s_{-i})$ iff $\forall s_i \in S_i$, $u_i(s_i^*, s_{-i}) \geq u_i(s_i, s_{-i})$
- Nash equilibrium:
  - $s = \langle s_1, \ldots, s_n \rangle$ is a Nash equilibrium iff $\forall i$, $s_i \in BR(s_{-i})$
- Every finite game has a Nash equilibrium! [Nash, 1950]
  - e.g., matching pennies: both players play heads/tails 50%/50%

Computing Mixed Nash Equilibria: Battle of the Sexes

	B	F
B	2, 1	0, 0
F	0, 0	1, 2

- It's hard in general to compute Nash equilibria, but it's easy when you can guess the support
- For BoS, let's look for an equilibrium where all actions are part of the support

Computing Mixed Nash Equilibria: Battle of the Sexes

	B	F
B	$2, 1$	$0, 0$
F	$0, 0$	$1, 2$

- Let player 2 play B with p, F with $1 - p$.
- If player 1 best-responds with a mixed strategy, player 2 must make him indifferent between F and B (why?)

Computing Mixed Nash Equilibria: Battle of the Sexes

	B	F
B	$2, 1$	$0, 0$
F	$0, 0$	$1, 2$

- Let player 2 play B with p, F with $1 - p$.
- If player 1 best-responds with a mixed strategy, player 2 must make him indifferent between F and B (why?)

$$u_1(B) = u_1(F)$$
$$2p + 0(1 - p) = 0p + 1(1 - p)$$
$$p = \frac{1}{3}$$

Computing Mixed Nash Equilibria: Battle of the Sexes

	B	F
B	$2, 1$	$0, 0$
F	$0, 0$	$1, 2$

- Likewise, player 1 must randomize to make player 2 indifferent.
  - Why is player 1 willing to randomize?

Computing Mixed Nash Equilibria: Battle of the Sexes

	B	F
B	$2, 1$	$0, 0$
F	$0, 0$	$1, 2$

- Likewise, player 1 must randomize to make player 2 indifferent.
  - Why is player 1 willing to randomize?

$$u_2(B) = u_2(F)$$
$$q + 0(1 - q) = 0q + 2(1 - q)$$
$$q = \frac{2}{3}$$

Computing Mixed Nash Equilibria: Battle of the Sexes

	B	F
B	$2, 1$	$0, 0$
F	$0, 0$	$1, 2$

- Thus the mixed strategies $(\frac{2}{3}, \frac{1}{3})$, $(\frac{1}{3}, \frac{2}{3})$ are a Nash equilibrium.

Interpreting Mixed Strategy Equilibria

What does it mean to play a mixed strategy? Different interpretations:

- Randomize to confuse your opponent
  - consider the matching pennies example
- Players randomize when they are uncertain about the other's action
  - consider battle of the sexes
- Mixed strategies are a concise description of what might happen in repeated play: count of pure strategies in the limit
- Mixed strategies describe population dynamics: 2 agents chosen from a population, all having deterministic strategies. MS is the probability of getting an agent who will play one PS or another.

Maxmin Strategies

- Player i's maxmin strategy is a strategy that maximizes i's worst-case payoff, in the situation where all the other players (whom we denote $-i$) happen to play the strategies which cause the greatest harm to i.
- The maxmin value (or safety level) of the game for player i is that minimum amount of payoff guaranteed by a maxmin strategy.

Definition (Maxmin)

The maxmin strategy for player i is $\arg\max_{s_i} \min_{s_{-i}} u_i(s_1, s_2)$, and the maxmin value for player i is $\max_{s_i} \min_{s_{-i}} u_i(s_1, s_2)$.

Minmax Strategies

- Player i's minmax strategy against player $-i$ in a 2-player game is a strategy that minimizes $-i$'s best-case payoff, and the minmax value for i against $-i$ is payoff.
- Why would i want to play a minmax strategy?

Definition (Minmax, 2-player)

In a two-player game, the minmax strategy for player i against player $-i$ is $\arg\min_{s_i} \max_{s_{-i}} u_{-i}(s_i, s_{-i})$, and player $-i$'s minmax value is $\min_{s_i} \max_{s_{-i}} u_{-i}(s_i, s_{-i})$.

Minmax Strategies

- Player i's minmax strategy against player $-i$ in a 2-player game is a strategy that minimizes $-i$'s best-case payoff, and the minmax value for i against $-i$ is payoff.
- Why would i want to play a minmax strategy?
  - to punish the other agent as much as possible

We can generalize to n players.

Definition (Minmax, n-player)

In an n-player game, the minmax strategy for player i against player $j \neq i$ is i's component of the mixed strategy profile s_{-j} in the expression $\arg\min_{s_{-j}} \max_{s_j} u_j(s_j, s_{-j})$, where $-j$ denotes the set of players other than j. As before, the minmax value for player j is $\min_{s_{-j}} \max_{s_j} u_j(s_j, s_{-j})$.

Minmax Theorem

Theorem (Minimax theorem (von Neumann, 1928))

In any finite, two-player, zero-sum game, in any Nash equilibrium each player receives a payoff that is equal to both his maxmin value and his minmax value.

1. Each player's maxmin value is equal to his minmax value. By convention, the maxmin value for player 1 is called the value of the game.
2. For both players, the set of maxmin strategies coincides with the set of minmax strategies.
3. Any maxmin strategy profile (or, equivalently, minmax strategy profile) is a Nash equilibrium. Furthermore, these are all the Nash equilibria. Consequently, all Nash equilibria have the same payoff vector (namely, those in which player 1 gets the value of the game).

Domination

- Let s_i and s'_i be two strategies for player i, and let S_{-i} be is the set of all possible strategy profiles for the other players

Definition

s_i strictly dominates s'_i if $\forall s_{-i} \in S_{-i}$, $u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i})$

Definition

s_i weakly dominates s'_i if $\forall s_{-i} \in S_{-i}$, $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$ and $\exists s_{-i} \in S_{-i}$, $u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i})$

Definition

s_i very weakly dominates s'_i if $\forall s_{-i} \in S_{-i}$, $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$

Equilibria and Dominance

- If one strategy dominates all others, we say it is dominant.
- A strategy profile consisting of dominant strategies for every player must be a Nash equilibrium.
  - An equilibrium in strictly dominant strategies must be unique.

Equilibria and Dominance

- If one strategy dominates all others, we say it is dominant.
- A strategy profile consisting of dominant strategies for every player must be a Nash equilibrium.
  - An equilibrium in strictly dominant strategies must be unique.
- Consider Prisoner's Dilemma again
  - not only is the only equilibrium the only non-Pareto-optimal outcome, but it's also an equilibrium in strictly dominant strategies!

Dominated Strategies

- No equilibrium can involve a strictly dominated strategy
  - Thus we can remove it, and end up with a strategically equivalent game
  - This might allow us to remove another strategy that wasn't dominated before
  - Running this process to termination is called iterated removal of dominated strategies.

Iterated Removal of Dominated Strategies: Example

	L	C	R
U	$3, 1$	$0, 1$	$0, 0$
M	$1, 1$	$1, 1$	$5, 0$
D	$0, 1$	$4, 1$	$0, 0$

Iterated Removal of Dominated Strategies: Example

	L	C	R
U	$3, 1$	$0, 1$	$0, 0$
M	$1, 1$	$1, 1$	$5, 0$
D	$0, 1$	$4, 1$	$0, 0$

- R is dominated by L.

Iterated Removal of Dominated Strategies: Example

	L	C
U	$3, 1$	$0, 1$
M	$1, 1$	$1, 1$
D	$0, 1$	$4, 1$

Iterated Removal of Dominated Strategies: Example

	L	C
U	$3, 1$	$0, 1$
M	$1, 1$	$1, 1$
D	$0, 1$	$4, 1$

- M is dominated by the mixed strategy that selects U and D with equal probability.

Iterated Removal of Dominated Strategies: Example

	L	C
U	3, 1	0, 1
D	0, 1	4, 1

Iterated Removal of Dominated Strategies

- This process preserves Nash equilibria.
  - strict dominance: all equilibria preserved.
  - weak or very weak dominance: at least one equilibrium preserved.
- Thus, it can be used as a preprocessing step before computing an equilibrium
  - Some games are solvable using this technique.
  - Example: Traveler's Dilemma!
- What about the order of removal when there are multiple dominated strategies?
  - strict dominance: doesn't matter.
  - weak or very weak dominance: can affect which equilibria are preserved.